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United States Patent Application Publication

20250273370

Kind Code

A1

Publication Date

August 28, 2025

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MAGNETIC POSITIONING HOLDER

Abstract

The present invention discloses a magnetic positioning holder for positioning a cylindrical storage container, comprising a main body and two magnetic strips. The main body features two guide surfaces on one side in the thickness direction, each adjacent to one of the sides of the main body, wherein each guide surface is concave toward the other side of the main body in the thickness direction to form an arc. Each magnetic strip is embedded in the main body and corresponds to each guide surface along the thickness direction of the main body. The distance between each magnetic strip and its corresponding guide surface, defined as the magnetic attraction distance, decreases from each width side toward the central part of the main body along the width direction. Thus, the storage container can be guided to move toward the central part of the main body in the width direction and positioned.

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Family ID: 1000007832267

Appl. No.: 18/585025

Filed: February 22, 2024

Publication Classification

Int. Cl.: H01F7/02 (20060101)

U.S. Cl.:

CPC H01F7/0252 (20130101);

Background/Summary

FIELD OF INVENTION

[0001] The present invention relates to a tool for positioning storage containers, in particular to a magnetic positioning holder.

BACKGROUND OF THE INVENTION

[0002] The assembly or maintenance of equipment often requires different types or specifications of tools. Throughout the assembly or maintenance process, the tools, components to be installed onto the equipment, or the components to be removed from the equipment can be selectively stored in a storage container. This container is strategically placed within easy reach of the operator to enhance the convenience of the assembly or maintenance operation.

[0003] When at least a part of the storage container is made of a metal material that can be magnetically attracted, a magnetic holder is commonly used to position the storage container. As shown in FIG. 1, the magnetic holder comprises a main body **11** having a long plate-like structure and multiple magnets **12**. The main body **11** forms a surface **13** on one side in the thickness direction, and the magnets **12** are spaced along the length direction inside the main body **11**. Each magnet **12** is positioned relative to the center of the plane **13** in the width direction.

[0004] A facade (not shown) could be provided on the side relative to another side of the surface **13** of the main body **11** in the thickness direction made of a metal material that can be magnetically attracted. This facade could be, for example, a side panel of a tool cabinet. By leveraging the magnetic force of each magnet **12** to position the main body **11** vertically, the storage container can then be positioned laterally against the surface **13**, with each magnet **12** attracting the container to ensure its placement and securement.

[0005] The external configuration of the storage container is not specifically limited; it may be a rectangular cylinder, a cylindrical shape, or any other shape. However, for non-planar cylindrical or other shaped containers, when the container is placed against the surface **13**, the container makes linear contact with the surface **13**. When the storage container is then placed on the main body **11**, the operator may inadvertently misalign it by failing to accurately position the container in the central part where each magnet **12** is located in the width direction of the surface **13**. This oversight or other factors may compromise the stability of the storage container positioning.

SUMMARY OF THE INVENTION

[0006] The main purpose of the present invention is to provide a magnetic positioning holder.

[0007] To achieve this purpose, the present invention adopts the following technical solution:

[0008] The magnetic positioning holder used for magnetically positioning a cylindrical storage container, at least a part of which is made of a metal material that can be magnetically attracted.

[0009] The magnetic positioning holder comprises:

[0010] A main body having a long plate-like structure. The main body features an upper edge side and a lower edge side at each end of its longitudinal direction, width sides formed on both sides in the width direction, and two guide surfaces formed on one side in the thickness direction. Each guide surface is adjacent to each width side and is concave toward the other side of the main body in the thickness direction to form an arc. The arcuate extension direction of each guide surface is directed toward the adjacent width side.

[0011] Two magnetic strips, each being a longitudinally magnetized rectangular strip. Each magnetic strip is embedded in the main body for magnetically attracting the storage container. The magnetic strips are parallel to each other, and the longitudinal direction of each is directed toward the upper edge side and the lower edge side. Each magnetic strip is adjacent to each width side, and each magnetic strip corresponds to each guide surface along the thickness direction of the main body.

[0012] The distance between each magnetic strip and its corresponding guide surface along the thickness direction of the main body is defined as the magnetic attraction distance. Along the width direction of the main body, the magnetic attraction distance decreases from each width side toward

the central part of the main body. Thus, the storage container can be guided to move toward the central part of the main body in the width direction and positioned.

[0013] The main effect and advantage of the present invention are to guide and position the storage container toward the central portion of the main body in the width direction, facilitating the collective magnetic adsorption positioning of the magnetic strips, thereby increasing the stability of the storage container positioning.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a three-dimensional view of a magnetic holder of the prior art;

[0015] FIG. 2 is a three-dimensional view of a preferred embodiment of the present invention;

[0016] FIG. 3 is a front elevational view of a preferred embodiment of the present invention;

[0017] FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 3;

[0018] FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 3;

[0019] FIG. 6 is a cross-sectional view taken along line 6-6 of FIG. 3;

[0020] FIG. 7 is a three-dimensional view of the state of use of a preferred embodiment of the present invention;

[0021] FIG. 8 is a partial cross-sectional view of the state of use of a preferred embodiment of the present invention;

[0022] FIG. 9 is a partial cross-sectional view of the state of use of a preferred embodiment of the present invention; and

[0023] FIG. 10 is a cross-sectional view of another state of use of a preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0024] The drawings herein depict a preferred embodiment of a magnetic positioning holder of the present invention. However, the following embodiments are for illustrative purposes only and do not limit the scope of the structure in the patent application.

[0025] As shown in FIGS. 2 to 10, the preferred embodiment of the magnetic positioning holder is used to magnetically position a cylindrical storage container 90, at least a part of which is made of a metal material that can be magnetically attracted.

[0026] The preferred embodiment of the magnetic positioning holder comprises a main body 20 having a long plate-like structure and four magnetic strips 30. The main body 20 features an upper edge side 21 and a lower edge side 22 at each end of its longitudinal direction, width sides 23 formed on both sides in the width direction, and two guide surfaces 42 formed on one side in the thickness direction. Each guide surface 42 is adjacent to each width side 23 and is concave toward the other side of the main body 20 in the thickness direction to form an arc. The arcuate extension direction of each guide surface 42 is directed toward the adjacent width side 23. One end of each guide surface 42 selectively extends toward the upper edge side 21, and the other end extends toward the lower edge side 22. In this embodiment, the end of each guide surface 42 is away from the upper edge side 21 and selectively closer to the lower edge side 22, but does not reach to the lower edge side 22.

[0027] Each magnetic strip 30 is a longitudinally magnetized rectangular strip. Each magnetic strip 30 is selectively embedded in the main body 20 from the rear side for magnetically attracting the storage container 90. The magnetic strips 30 are parallel to each other, and the longitudinal direction of each is directed toward the upper edge side 21 and the lower edge side 22. Each magnetic strip 30 is adjacent to each width side 23, and each magnetic strip 30 corresponds to each guide surface 42 along the thickness direction of the main body 20.

[0028] The magnetic strips 30 are arranged in pairs in a side-by-side configuration, and their

number may vary based on the length of the main body **20**, but is guided by the principle of having an even number of magnetic strips **30**, with a minimum of two. The main body **20** forms multiple passages **26** on the rear side, each of which provides a space for each magnetic strip **30** to be embedded. When the main body **20** is made of a material with elastic deformability, the length of each passage **26** is slightly shorter than the length of each magnetic strip **30**. When the magnetic strips **30** are embedded in the main body **20**, the magnetic strips **30** can pass through the passages **26** by bending and deforming the main body **20**.

[0029] The distance between each magnetic strip **30** and its corresponding guide surface **42** along the thickness direction of the main body **20** is defined as a magnetic attraction distance D . Along the width direction of the main body **20**, the magnetic attraction distance D decreases from each width side **23** toward the central part of the main body **20**. Thus, the storage container **90** can be guided to move toward the central part of the main body **20** in the width direction and positioned.

[0030] The main body **20** forms a connecting surface **44** on one side in the thickness direction, which is concave toward the other side of the main body **20** to form an arc. The connecting surface **44** is located between each guide surface **42**, and together, each guide surface **42** and the connecting surface **44** form an arcuate surface **40**.

[0031] FIGS. **7** and **8** illustrate a preferred embodiment in use for positioning a cylindrical storage container **90**. The outer periphery of the storage container **90** features a circular outer surface **92** that matches the diameter of the arcuate surface **40**. At this point, the outer surface **92** forms a face contact when attached to the arcuate surface **40**, which significantly increases the stability of positioning the storage container **90** as compared to the prior art.

[0032] FIG. **10** illustrates the state of use in which the outer diameter of the outer surface **92** is greater than the diameter of the arcuate surface **40**. The formation of the arcuate surface **40** results in the outer surface **92** forming two line contacts with the main body **20**. This further increases the stability of the positioning of the storage container **90** as compared to the prior art.

[0033] Furthermore, the formation of each guide surface **42** causes the magnetic attraction distance D to decrease from each width side **23** toward the central part of the main body **20**. The magnetic effect of each magnetic strip **30** on the storage container **90** when it is near the central part of the main body **20** is greater than the magnetic effect when it is away from the central part of the main body **20**. This difference in magnetic effects can guide the storage container **90** to move toward the central part of the main body **20** in the width direction, and to be positioned when the storage container **90** is inadvertently placed on either side of the width side **23**. This is advantageous for the common magnetic adsorption positioning of each magnetic strip **30**, thereby increasing the stability of the positioning of the storage container **90**.

[0034] A support **24** protrudes from the front side of the main body **20** and extends to the lower edge side **22**. As shown in FIG. **9**, the support **24** supports the storage container **90**, thereby increasing the stability of the positioning of the storage container **90**.

[0035] A gap **25** is formed between the support **24** along the positive projection of the main body **20** in the longitudinal direction and the arcuate surface **40**. This selectively allows the storage container **90** to be pressed downward when the storage container **90** has a protruding annular convex ring **94** at the bottom, and when the storage container **90** is magnetically adsorbed by the magnetic strips **30**, the convex ring **94** can be pressed downward to enter the gap **25**, and the inner side of the support **24** corresponding to the annular convex ring **94** can form a lateral constraint on the convex ring **94**. This prevents the storage container **90** from easily moving away from the arcuate surface **40** in the thickness direction of the main body **20**, and the storage container **90** is less likely to be detached from the main body **20**, thereby increasing the stability of the positioning of the storage container **90**.

Claims

1. A magnetic positioning holder for magnetically positioning a cylindrical storage container, at least a part of which is made of a metal material that can be magnetically attracted; the magnetic positioning holder comprises: a main body with a long plate-like structure featuring an upper edge side and a lower edge side at both ends in the longitudinal direction, width sides formed on both sides in the width direction, and two guide surfaces formed on one side in the thickness direction, wherein each guide surface is adjacent to each width side and is concave toward the other side of the main body in the thickness direction to form an arc, and the arcuate extension direction of each guide surface is directed toward the adjacent width side; and two magnetic strips, each being a longitudinally magnetized rectangular strip, embedded in the main body for magnetically attracting the storage container, wherein the magnetic strips are parallel to each other and the longitudinal direction of each magnetic strip is directed toward the upper edge side and the lower edge side, each magnetic strip is adjacent to each width side, and each magnetic strip corresponds to each guide surface along the thickness direction of the main body; wherein the distance between each magnetic strip and its corresponding guide surface along the thickness direction of the main body is defined as the magnetic attraction distance, and along the width direction of the main body, the magnetic attraction distance decreases from each width side toward the central part of the main body, thereby guiding the storage container to move toward and be positioned at the central part of the main body in the width direction.

2. The magnetic positioning holder according to claim 1, wherein the main body forms a connecting surface on one side in the thickness direction, and the connecting surface is concave toward the other side of the main body to form an arc; the connecting surface is located between each guide surface, and each guide surface and the connecting surface together form an arcuate surface.

3. The magnetic positioning holder according to claim 1, wherein a support protrudes from the front side of the main body and extends to the lower edge side used for supporting the storage container.

4. The magnetic positioning holder according to claim 2, wherein a support protrudes from the front side of the main body and extends to the lower edge side used for supporting the storage container; a gap is formed between the support along the positive projection of the main body in the longitudinal direction and the arcuate surface, allowing a convex ring at the bottom of the storage container to enter the gap.
